

8086 Conditional Jump Instructions

1. Introduction

Concept

Conditional jump instructions transfer program control based on the result of a condition test on status flags.

Rules / Notes

Key Characteristics

- All conditional jumps are **short jumps**
- Jump range: **+127 bytes to -128 bytes**
- Conditions are tested using flag bits:

CF, ZF, SF, OF, PF

Rules / Notes

Execution Logic

- If condition is **true** → control jumps to label
- If condition is **false** → next instruction executes

2. Signed vs Unsigned Comparisons

Concept

8086 provides two logical groups of conditional jumps, depending on whether numbers are treated as **signed** or **unsigned**.

Rules / Notes

Unsigned Comparisons

- Use **CF** and **ZF**
- Instructions: **JA, JAE, JB, JBE, JE, JNE**

Signed Comparisons

- Use **SF** and **OF**
- Instructions: **JG, JGE, JL, JLE, JE, JNE**

Exam Tip

Exam Rule: Unsigned → Carry flag (CF) Signed → Sign flag (SF) and Overflow flag (OF)

3. Conditional Jump Instruction Table

Instruction	Tested Condition	Meaning
JA	ZF=0 and CF=0	Jump if above (unsigned)
JAЕ	CF=0	Jump if above or equal
JB	CF=1	Jump if below
JBE	ZF=1 or CF=1	Jump if below or equal
JC	CF=1	Jump if carry
JNC	CF=0	Jump if no carry
JE / JZ	ZF=1	Jump if equal / zero
JNE / JNZ	ZF=0	Jump if not equal / not zero
JG	ZF=0 and SF=OF	Jump if greater (signed)
JGE	SF=OF	Jump if greater or equal
JL	SF≠OF	Jump if less
JLE	ZF=1 or SF≠OF	Jump if less or equal
JO	OF=1	Jump if overflow
JNO	OF=0	Jump if no overflow
JS	SF=1	Jump if sign (negative)
JNS	SF=0	Jump if no sign (positive)
JP / JPE	PF=1	Jump if parity even
JNP / JPO	PF=0	Jump if parity odd
JCXZ	CX=0	Jump if CX is zero

4. Example: Conditional Jump with String Instruction

Example

```
; SCASB searches a table of 100 bytes for 0AH
; Address of TABLE is assumed to be in ES:DI

MOV CX, 100          ; load counter
MOV AL, 0AH          ; byte to search
CLD                  ; auto-increment DI
REPNE SCASB          ; repeat while not equal and CX != 0

JCXZ NOT_FOUND       ; jump if CX = 0 (not found)
STC                  ; set carry flag if found

NOT_FOUND:
```

Explanation:

- REPNE SCASB scans memory until AL matches or CX becomes zero
- If CX reaches zero → value not found
- JCXZ jumps when CX = 0
- If found, carry flag is set using STC

Quick Revision

- Conditional jumps depend on flag values
- All conditional jumps are short (± 128 bytes)
- Signed comparisons use SF and OF
- Unsigned comparisons use CF
- JZ/JNZ test ZF directly
- JCXZ is independent of flags